

ChatGPT said:

બિલકુલ. પહેલા હું તમને 1A થી 2D સુધીનું સંપૂર્ણ Professional, Exam-Ready version આપું છું.
ત્યારબાદ તમે વાંચીને જે કંઈ કમી/ફેરફાર જોઈએ તે કહેજો.

DIGITAL COMMUNICATION – EXAM NOTES

SECTION – Q.1

Total Marks: 10 | Each Question: 05 Marks

Q.1 (A) Draw and Explain Block Diagram of Digital Communication System

[05 Marks]

Block Diagram

Information

Source /

Transducer

|

v

+-----+

| Source Encoder |

+-----+

|

v

+-----+

| Channel Encoder |

+-----+

|

v

+-----+

| Modulator |

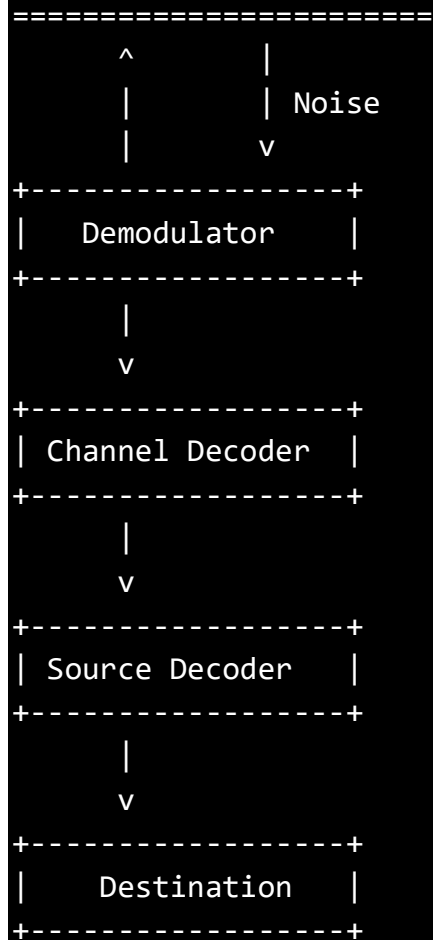
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|

v

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CHANNEL



Explanation of Blocks

1. Information Source / Input Transducer:

The information source generates the message such as **speech, image, video or computer data**. The input transducer converts a non-electrical message into an electrical signal, if required.

2. Source Encoder:

It removes unnecessary/redundant information from the message and represents it using minimum number of bits. It also performs **data compression**.

Example: Huffman coding.

3. Channel Encoder:

It adds controlled redundant bits to the data so that errors introduced by noise during transmission can be **detected and/or corrected** at the receiver.

4. Modulator:

The digital signal is converted into a suitable signal for transmission. In digital communication, the carrier parameter such as **amplitude, frequency or phase** is varied according to the digital data.

5. Channel:

It is the physical medium through which the signal travels.

Examples: Air, coaxial cable, optical fibre.

During transmission, unwanted **noise and interference** may affect the signal.

6. Demodulator:

It receives the modulated signal and recovers the transmitted digital information from it.

7. Channel Decoder:

It detects and corrects errors introduced during transmission and recovers the original encoded data.

8. Source Decoder:

It decompresses/decodes the received data and reconstructs the original information.

9. Destination:

The destination is the final user or device that receives the reconstructed information.

In Short

Source → Source Encoder → Channel Encoder → Modulator
→ Channel → Demodulator → Channel Decoder
→ Source Decoder → Destination

Q.1 (B) Write Electromagnetic Wave Spectrum, Frequency Bands and Their Applications

[05 Marks]

Definition

The **Electromagnetic (EM) Spectrum** is the complete range of electromagnetic radiation arranged according to their **frequency or wavelength**.

The relationship between frequency and wavelength is:

$$c=f\lambda$$

Where,

- $c=3 \times 10^8$ m/s = velocity of electromagnetic wave
- f = frequency
- λ = wavelength

Electromagnetic Frequency Bands

Band	Frequency Range	Approx. Wavelength	Applications
ELF	3–30 Hz*	Very long	Power systems, geophysical studies
LF	30–300 kHz	100–10 km	Navigation, submarine communication
MF	300–3 MHz	100–100 m	Navigation, RFID, time signals
HF	3–30 MHz	10–100 m	AM radio broadcasting
VHF	30–300 MHz	1–10 m	Shortwave, amateur radio, aviation/marine communication
UHF	300–3 GHz	10–10 cm	FM radio, TV, aviation, marine communication
SHF	3–30 GHz	10–10 cm	Cellular, Wi-Fi, TV, GPS, radar
EHF	30–300 GHz	1–10 cm	Satellite communication, radar, microwave links
THF	300–3000 GHz	0.1–0.1 mm	Millimetre-wave communication, radar, remote sensing

*Some textbooks use different lower-frequency boundaries for ELF. Follow the classification specified by your syllabus if your college provides one.

Spectrum Representation

Frequency →

ELF | VLF | LF | MF | HF | VHF | UHF | SHF | EHF

Low Frequency
Long Wavelength

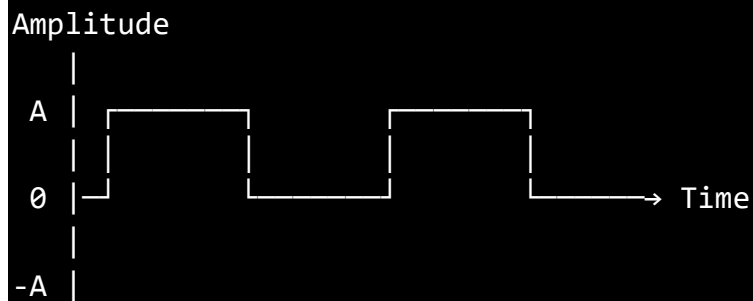
High Frequency
Short Wavelength

Q.1 (C) Draw Following Waveforms in Time Domain and Frequency Domain

[05 Marks]

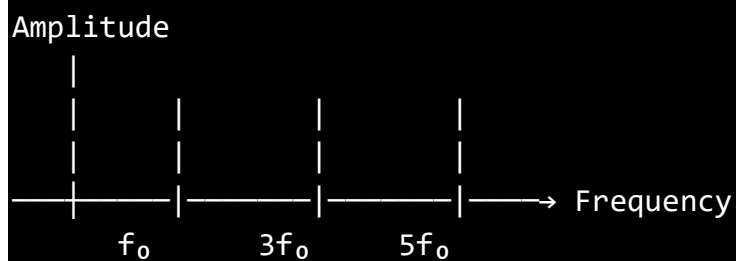
(i) Square Wave

Time Domain



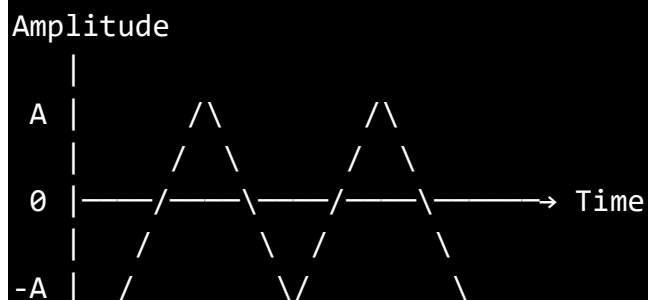
Frequency Domain

A square wave contains the **fundamental frequency and odd harmonics**.



(ii) Triangular Wave

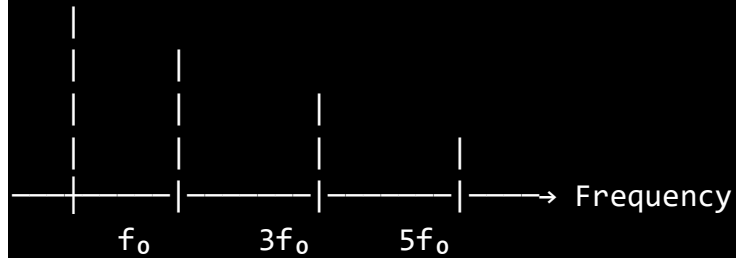
Time Domain



Frequency Domain

A triangular wave contains the **fundamental and odd harmonics**, with harmonic amplitudes decreasing rapidly.

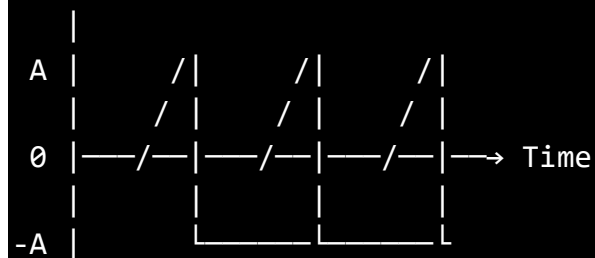
Amplitude



(iii) Sawtooth Wave

Time Domain

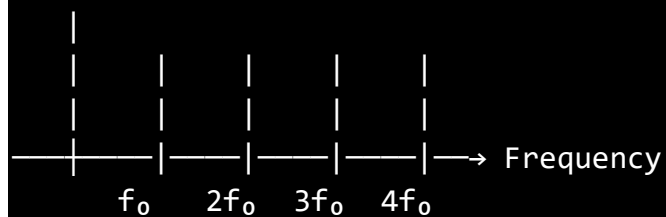
Amplitude



Frequency Domain

A sawtooth wave contains the **fundamental and both even and odd harmonics**.

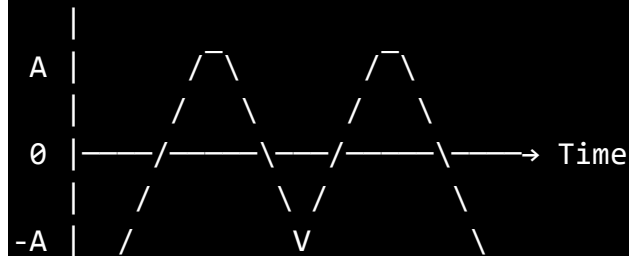
Amplitude



(iv) Sinusoidal Wave

Time Domain

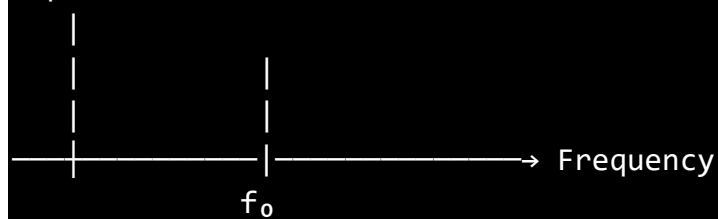
Amplitude



Frequency Domain

An ideal sinusoidal wave contains only **one frequency component**.

Amplitude



Important Point

- **Square wave:** Odd harmonics
- **Triangular wave:** Odd harmonics
- **Sawtooth wave:** Even + odd harmonics
- **Sinusoidal wave:** Only fundamental frequency

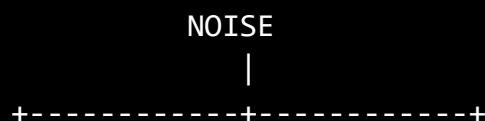
Q.1 (D) What is Noise? Classify Noise and Describe Each One

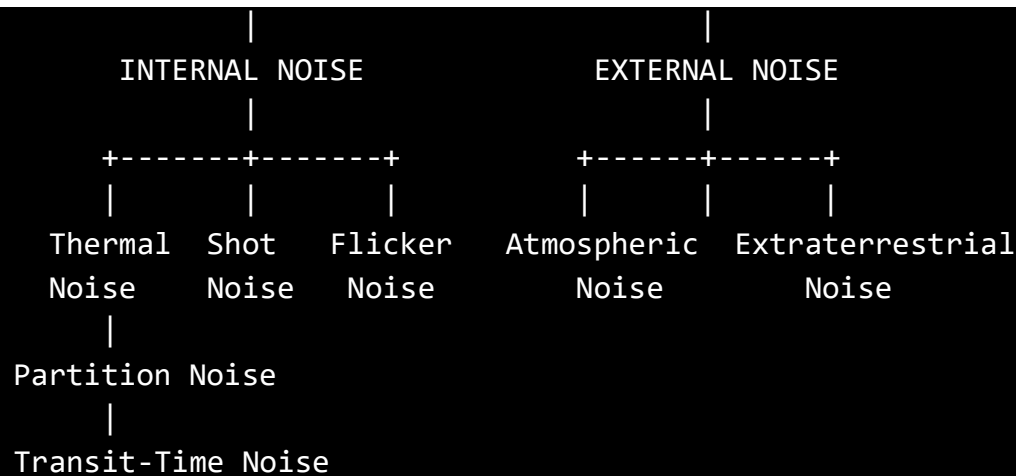
[05 Marks]

Definition of Noise

Noise is an unwanted random electrical signal that gets added to the desired signal during communication and causes **distortion or degradation** of the received signal.

Classification of Noise





1. Internal Noise

Internal noise is generated **within the electronic components** of the communication system.

(a) Thermal Noise

Thermal noise is caused by the **random motion of electrons due to temperature** in resistors and conductors.

It is also called **Johnson or Johnson-Nyquist noise**.

(b) Shot Noise

Shot noise occurs due to the **random flow of charge carriers** across semiconductor junctions such as diodes and transistors.

(c) Flicker Noise

Flicker noise is mainly a **low-frequency noise** present in electronic devices.

It is also known as **$1/f$ noise**.

(d) Partition Noise

Partition noise occurs when current is divided randomly between **two or more paths** in an electronic device.

(e) Transit-Time Noise

Transit-time noise becomes significant at **high frequencies**, when the carrier transit time becomes comparable to the signal period.

2. External Noise

External noise originates **outside the communication system**.

(a) Atmospheric Noise

It is produced by natural electrical disturbances such as:

- Lightning
- Thunderstorms
- Electrical storms

(b) Extraterrestrial Noise

It originates from sources outside the Earth.

Examples:

- Solar radiation
- Cosmic radiation
- Stars and galaxies

(c) Man-Made Noise

It is produced by human-made electrical and electronic equipment.

Examples:

- Automobile ignition systems
- Electric motors
- Power lines
- Industrial machinery

SECTION – Q.2

Total Marks: 10 | Each Question: 05 Marks

Q.2 (A) Define:

(i) Sensitivity (ii) Selectivity (iii) Fidelity (iv) Image Frequency Rejection

[05 Marks]

(i) Sensitivity

Sensitivity is the ability of a receiver to receive and amplify a **very weak signal** satisfactorily.

A receiver having high sensitivity can receive weaker signals.

Mainly depends on: Receiver gain and noise performance.

(ii) Selectivity

Selectivity is the ability of a receiver to select the **desired frequency/station** and reject unwanted signals or adjacent channels.

Mainly depends on: IF amplifier/filter characteristics.

(iii) Fidelity

Fidelity is the ability of a communication receiver to reproduce the **original information signal accurately without distortion**.

A high-fidelity receiver reproduces the original audio signal with minimum distortion.

(iv) Image Frequency Rejection

In a superheterodyne receiver, **image frequency** is an unwanted frequency that produces the same intermediate frequency as the desired signal.

For the high-side local oscillator:

$$f_o = f_s + f_{IF}$$

Image frequency:

$$f_{si} = f_s + 2f_{IF}$$

Image frequency rejection is the ability of the receiver's RF stage to reject this unwanted image frequency.

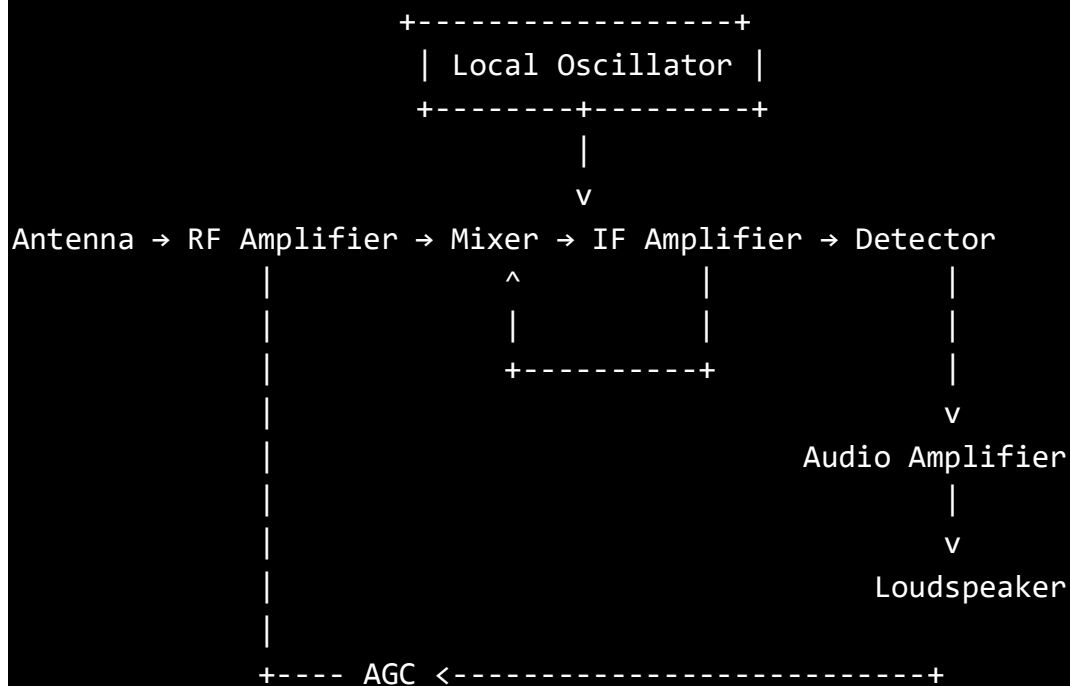
Important Point

Image-frequency rejection is mainly provided by the **RF tuned circuit/stage before the mixer**.

Q.2 (B) Draw and Explain Block Diagram of Superheterodyne AM Receiver

[05 Marks]

Block Diagram



Working

1. Antenna:

Receives the electromagnetic radio signal from space and converts it into an electrical signal.

2. RF Amplifier:

Selects the desired station, provides initial amplification and helps in **image-frequency rejection**.

3. Local Oscillator:

Generates a frequency f_o such that:

$$fIF = |f_o - f_s|$$

For standard AM broadcast receivers:

$$fIF = 455 \text{ kHz}$$

4. Mixer:

The received RF signal and local oscillator signal are mixed to produce the fixed intermediate frequency (IF).

$$fIF = |f_o - f_s|$$

5. IF Amplifier:

Provides most of the receiver's **gain and selectivity** at the constant IF frequency.

6. Detector:

Demodulates the AM signal and recovers the original audio/message signal. A **diode envelope detector** is commonly used.

7. Audio Amplifier:

Amplifies the recovered audio signal to a suitable power level.

8. Loudspeaker:

Converts the electrical audio signal into sound.

9. AGC (Automatic Gain Control):

Automatically adjusts receiver gain according to received signal strength to maintain a relatively constant output level.

Q.2 (C) Derive Mathematical Equation of Amplitude Modulated Wave

[05 Marks]

Let the carrier signal be:

$$e_c = E_c \sin \omega_c t$$

Where:

- E_c = Carrier amplitude

- ω_c = Carrier angular frequency

Let the modulating signal be:

$$e_m = E_m \sin \omega_m t$$

Where:

- E_m = Modulating signal amplitude
- ω_m = Modulating angular frequency

Step 1: Instantaneous Amplitude

In AM, the amplitude of the carrier varies according to the modulating signal.

Therefore,

$$A = E_c + E_m \sin \omega_m t$$

Step 2: AM Wave Equation

$$e_{AM} = A \sin \omega_c t$$

Therefore,

$$e_{AM} = (E_c + E_m \sin \omega_m t) \sin \omega_c t$$

Taking E_c common:

$$e_{AM} = E_c (1 + m \sin \omega_m t) \sin \omega_c t$$

The modulation index is:

$$m = \frac{E_m}{E_c}$$

Therefore,

$$e_{AM} = E_c (1 + m \sin \omega_m t) \sin \omega_c t$$

Step 3: Expanding the Equation

$$e_{AM} = E_c \sin \omega_c t + m E_c \sin \omega_m t \sin \omega_c t$$

Using:

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

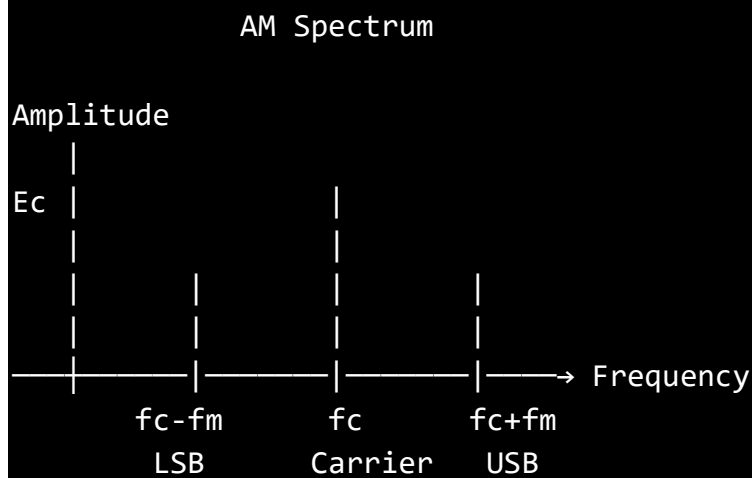
Therefore,

$$eAM = E_c \sin(\omega_c t) + m E_c \cos(\omega_c - \omega_m)t - m E_c \cos(\omega_c + \omega_m)t$$

Components of AM Wave

Component	Frequency	Amplitude
Carrier	f_c	E_c
Lower Sideband (LSB)	$f_c - f_m$	$\frac{1}{2} E_c m$
Upper Sideband (USB)	$f_c + f_m$	$\frac{1}{2} E_c m$

Final Result



Q.2 (D) What is Modulation? Why is it Needed? Draw and Explain Amplitude Modulation

[05 Marks]

Definition of Modulation

Modulation is the process of varying a parameter of a high-frequency carrier wave according to the instantaneous value of the low-frequency message signal.

The carrier parameter may be:

- Amplitude → **Amplitude Modulation (AM)**
- Frequency → **Frequency Modulation (FM)**
- Phase → **Phase Modulation (PM)**

Why Modulation is Needed?

1. Reduction in Antenna Size

For efficient transmission, antenna length is related to wavelength:

$$H \approx \lambda/4$$

Since:

$$\lambda = cf$$

Increasing the carrier frequency reduces wavelength and hence reduces the practical antenna size.

2. Avoids Mixing of Signals

Different message signals can be transmitted using different carrier frequencies, reducing interference between users/stations.

3. Enables Multiplexing

Modulation allows multiple signals to be transmitted through the same communication medium using different carrier frequencies.

4. Improves Transmission Efficiency

High-frequency carrier signals are more suitable for radiation and wireless transmission.

5. Reduces Interference

Proper carrier-frequency allocation and filtering help separate different transmitted signals.

Amplitude Modulation (AM)

Definition

Amplitude Modulation is the process in which the **amplitude of the carrier wave varies according to the instantaneous amplitude of the message signal**, while the carrier frequency and phase remain constant.

AM Waveforms

1. Message Signal

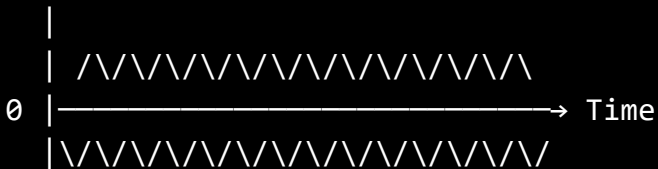
Amplitude

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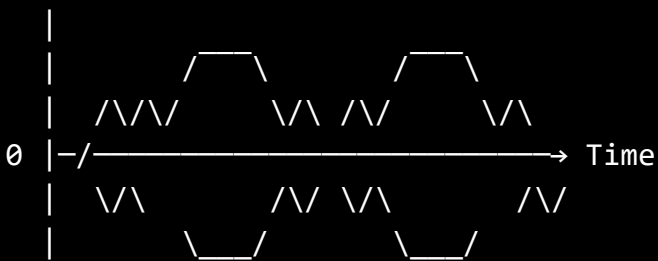
2. Carrier Signal

Amplitude



3. AM Signal

Amplitude



The envelope of the AM wave follows the message signal.

AM Equation

$$e_{AM} = E_c(1 + m \sin \omega_m t) \sin \omega_c t$$

Where:

$$m = \frac{E_m}{E_c}$$

For proper AM without over-modulation:

$$m \leq 1$$

For $m=1$, the modulation is **100%**.

★ QUICK REVISION – IMPORTANT FORMULAS

AM Wave

$$e_{AM} = E_c(1 + m \sin \omega_m t) \sin \omega_c t$$

Modulation Index

$$m = \frac{E_m}{E_c}$$

AM Sideband Frequencies

$$f_{USB} = f_c + f_m$$

$$f_{LSB} = f_c - f_m$$

Superheterodyne IF

$$f_{IF} = |f_o - f_s|$$

For standard AM receiver:

$$f_{IF} = 455 \text{ kHz}$$

Electromagnetic Wave Relation

$$c = f\lambda$$

Antenna Length

$$H \approx \lambda/4$$